

LA RIOJA, Argentina

WMO: 872170

Lat: 29.382S

Lon: 66.793W

Elev: 1407

StdP: 13.96

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.2	37.9	23.3	18.5	47.8	26.4	21.4	46.7	17.9	54.1	14.1	53.6	4.0	270	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.5	102.3	74.2	99.7	73.7	97.1	73.3	78.6	94.2	77.2	92.8	75.9	91.2	6.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	135.3	87.1	72.9	128.8	85.2	71.7	123.3	83.6	43.5	94.5	41.9	92.9	40.6	91.6	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.9	13.8	11.9	DB	29.5	108.1	3.2	2.3	27.1	109.8	25.3	111.1	23.5	112.4	21.1	114.1
			WB	27.1	81.9	3.3	2.3	24.8	83.6	22.8	84.9	21.0	86.2	18.6	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.6	82.3	80.0	76.0	68.5	61.1	54.7	53.7	59.6	66.7	73.9	78.3	81.5	(6)
	DBStd	11.95	5.73	6.33	6.24	6.59	6.37	5.30	5.61	7.16	7.81	7.56	6.83	6.19	(7)
	HDD50	55	0	0	0	1	3	17	26	7	1	0	0	0	(8)
	HDD65	1145	0	1	4	36	156	310	353	197	71	15	2	0	(9)
	CDD50	7225	1001	841	806	557	347	158	141	306	502	740	850	976	(10)
	CDD65	2839	536	422	345	142	35	1	2	31	122	290	402	511	(11)
	CDH74	31458	6361	4660	3108	1066	247	20	58	485	1325	3168	4615	6345	(12)
	CDH80	15878	3473	2392	1376	349	42	2	13	170	527	1538	2456	3540	(13)
(14) Wind	WSAvg	5.1	5.3	5.0	4.8	4.2	3.9	3.7	4.5	5.2	5.6	6.0	6.1	6.3	(14)
(15) Precipitation	PrecAvg	14.6	3.3	2.8	2.3	1.1	0.2	0.2	0.2	0.2	0.2	0.6	1.5	2.5	(15)
	PrecMax	25.2	9.0	8.0	8.1	7.6	1.1	1.3	1.1	2.1	1.1	2.9	7.7	8.6	(16)
	PrecMin	7.0	0.3	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	4.6	2.0	2.0	1.9	1.2	0.3	0.3	0.2	0.4	0.2	0.7	1.2	1.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.6	103.4	98.8	93.1	84.3	76.7	80.0	92.9	96.8	102.0	104.2	106.1	(19)
		MCWB	77.5	75.9	75.2	73.4	68.1	62.6	60.9	65.8	67.2	71.5	72.6	73.6	(20)
	2%	DB	101.2	98.9	94.9	87.8	80.1	72.1	74.1	84.1	90.4	96.9	99.6	101.6	(21)
		MCWB	76.2	75.2	73.7	71.1	66.4	59.3	58.7	62.5	65.7	71.1	71.4	73.7	(22)
	5%	DB	98.1	95.7	91.5	83.9	76.5	69.5	70.5	79.0	86.0	92.7	96.3	98.6	(23)
		MCWB	75.1	74.6	72.8	69.2	64.2	57.7	56.6	60.3	63.8	68.7	71.0	73.2	(24)
	10%	DB	94.7	92.5	87.8	80.2	72.6	66.5	66.9	74.5	81.7	88.7	92.6	95.3	(25)
		MCWB	74.2	73.9	71.7	67.3	62.2	56.2	54.5	57.9	61.3	67.0	69.8	72.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.2	80.0	78.6	75.4	70.1	64.7	62.9	67.3	70.6	75.9	76.2	78.7	(27)
		MCDB	99.6	94.4	93.8	89.0	80.1	75.3	76.4	88.0	92.4	95.7	96.7	97.6	(28)
	2%	WB	78.9	78.1	76.3	73.1	67.8	61.2	60.1	63.9	67.7	72.4	74.2	76.7	(29)
		MCDB	94.7	92.4	89.6	83.6	77.4	69.7	72.2	82.2	87.1	92.1	94.3	95.5	(30)
	5%	WB	77.3	76.8	74.6	71.1	65.7	59.0	57.6	61.3	65.0	70.3	72.5	75.3	(31)
		MCDB	92.6	91.2	86.5	80.3	74.2	67.1	68.7	77.6	82.6	89.3	91.4	93.3	(32)
	10%	WB	75.7	75.4	73.1	69.2	63.3	57.1	55.4	58.6	62.7	68.2	71.0	73.8	(33)
		MCDB	90.4	89.1	84.3	77.8	71.1	64.7	65.2	73.1	78.9	85.1	88.8	90.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	20.7	19.1	19.4	19.3	22.1	24.9	26.0	25.1	23.6	23.4	23.2	(35)
		MCDBR	25.9	25.2	24.5	24.5	24.1	26.6	30.3	32.8	30.1	28.4	28.3	28.3	(36)
	5% WB	MCWBR	9.3	9.3	9.3	11.6	13.1	15.9	17.6	16.9	14.3	12.6	11.5	10.6	(37)
		MCWBR	22.4	21.5	20.6	20.7	20.9	24.0	27.6	30.6	27.8	25.5	25.3	24.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.396	0.385	0.369	0.353	0.326	0.308	0.302	0.335	0.377	0.376	0.380	0.391	(40)
		taud	2.418	2.454	2.495	2.492	2.511	2.529	2.522	2.411	2.302	2.366	2.403	2.413	(41)
	Ebn at Noon		300	298	292	281	274	273	280	282	284	297	303	302	(42)
		Edh at Noon	40	37	34	30	27	25	26	32	40	40	40	40	(43)
(44) All-Sky Solar Radiation	RadAvg		2309	2066	1761	1420	1127	1048	1166	1482	1851	2174	2403	2428	(44)
	RadStd		120	141	117	115	104	76	59	75	113	135	121	130	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page